Activities 2016 / 2017

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Abstract

This document describes the university classes, GSSI classes, summer schools and seminars attended in the 2016 / 2017 academic year. The bibliography contains the list of the books and articles that mean the basis of this research project.

Contents

1 University of L'Aquila

1.1 Numerical Analysis 2 - Prof. Guglielmi

Euler method. Steepest descent gradient method. Conjugate directions method. Matlab.

1.2 High Performance Parallel Computing - several Professors

1.2.1 Numerical solutions of calculus of variations problems - Prof. Protasov

Let's consider the following problem, also known as The Simplest Problem. We aim to minimize the function $J(x) = \int_{t_0}^{t_1} L(t, x, \dot{x})dt$, where $t \in (t_0, t_1)$ and $x \in C^1[t_0, t_1]$. The most important equation we need when discussing this problem is the Euler-Lagrange equation, 1.

$$\frac{d}{dt} L_{\dot{x}} = L_x \tag{1}$$

It is proved that if $\hat{x}$ provides local minimum to The Simplest Problem, then it satisfies the Euler-Lagrange equation and several examples are shown. It is also highlighted that if L is convex, then $\hat{x}$ is necessarily an absolute minimum.

Three special cases of equation 1 are considered which are useful when solving problems.

1. L does not depend explicitly on $\dot{x}$, ie. $L = L(t, x)$. In this case equation 1 is reduced to $L_x = 0$.
2. L has the form of $L = L(t, \dot{x})$, and hence $L_{\dot{x}}$ is constant.
3. When L does not explicitly depend on the time variable t we have $\dot{x}L_{\dot{x}} - L$ being constant.

It is important to note that when a solution x is not smooth enough, ie. if it lives in a wider space than C^1, it can be rejected by the Euler-Lagrange method, however it is still a solution to The Simplest Problem.

As a variation of the original problem, the Bolza problem is introduced, in which we want to minimize $J(x) = \int_{t_0}^{t_1} L(t, x, \dot{x})dt + l(x_0, x_1)$, and accordingly, a similar theorem to the previous one is described for a $\hat{x}$ solution of the problem.

A good example for the many applications of The Simplest Problem is the solution of the Isoperimetric Problem in which it is shown it is the circle that has the biggest possible area among forms with concrete length. In the solution a pair of Lagrange multipliers are used and the theorem that had been shown.

As an interesting aspect, Jacob Steiner's intuitive but wrong version of the proof is described for the same Isoperimetric Problem from 1850 as to illustrate the significance of the Euler-Lagrange method.

The next problem we considered throughout the course of this lectur was the following boundary value problem.

$$\begin{cases} \ddot{x} = f(t, x, \dot{x}) \\ x(t_0) = x_0, x(t_1) = x_1 \end{cases}$$

The approach we were introduced is the so-called "shooting method". Instead of solving the above problem, we can consider the Cauchy problem.

$$\begin{cases} \ddot{x} = f(t, x, \dot{x}) \\ x(t_0) = x_0, \dot{x}(t_0) = v_0 \end{cases}$$

This Cauchy problem is basically the same as the more simple one, with first order derivatives, using the new notation $x_1 = x, x_2 = \dot{x}$. For the Cauchy problem there exists an existence and uniqueness theorem which says that if f is Lipschitz continuous, then we have a unique solution on an interval. Taking this approach we only have to connect somehow the solution of this new problem to the original BVP. This can be done by introducing a new variable, $\alpha \in \mathbb{R}$, and then we can consider several solutions ("shootings") $P(\alpha) = x_\alpha(t_1) - x_1$, and find $P(\alpha) = 0$ by the bisection, or double decision method.

Difference schemes and difference equations are also discussed. An equation is stable if and only if all roots of the characteristic polynomial are strictly smaller than 1 by modulus, and it is called α-stable is the polynomial has one root equal to 1 and all other roots are smaller than 1.

1.2.2 Optimization of complex systems - Prof. Shcherbatyy

This section of the HPC lecture introduces optimization problems of complex systems. Two motivational, illustrative applications are considered in the beginning of the course, partly to fix the terminology and introduce notions such as state function and control function. Let's consider the following problem: how to boil a potato by heating its boundary in an optimal way. To put in into mathematical forms, let Ω be the potato, Γ its boundary. Let $u(x)$ denote the heat source or the control function on Γ, and finally let $y(x)$ be the temperature, or in other words, the state function in Ω.

We have an elliptic boundary control problem

$$-\Delta y = 0, x \in \Omega$$
$$\frac{\partial y}{\partial n} = \alpha(u - y), x \in \Gamma,$$

and to express the objective we formulate

$$\min_{y,u} J(y, u) = \frac{1}{2} \int_\Omega (y(x) - y_d(x))^2 dx - \frac{\gamma}{2} \int_\gamma u(x)^2 ds(x),$$

while the following inequality constraints hold

$$u_a(x) \leq u(x) \leq u_b(x), x \in \Gamma.$$

A different version of this problem is when we have distributed control inside Ω.

Using finite differences or finite elements method we obtain a discrete problem of the following form

$$\min_{Y,U} \frac{1}{2} \sum_{i=1}^{N} (y_i - y_{d,i})^2 + \gamma u_i^2,$$

subject to $A_h Y = B_h U$ and $U_a \leq U \leq U_b$, where $Y, U \in \mathbb{R}^N$ and $A_h, B_h \in \mathbb{R}^{N \times N}$. The general solution methods contain Newton method, steepest descent gradient method and conjugate gradient method, which were elaborated also in Numerical Analysis 2.

The problem we considered above has application in distributed heating in medicine.

Finally, the topic of sensitivity analysis was introduced, which, roughly speaking, calculates the rates of change in the output variables of a system which result from small perturbations in the problem parameters. Direct differentiation method and adjoint method were considered in detail.

1.2.3 Numerical solution of Boundary value problems for PDEs - Prof. Khapko

1.2.4 Programming and using Caliban - Prof. Pera

Basic Linux commands. Practical part, actual parallel computing.

2 Gran Sasso Science Institute

2.1 Introduction to Finite Elements Method - Prof. Boffi

Finite differences, finite elements.

2.2 Decay property for hyperbolic systems with dissipation - Prof. Kawashima

The aim of the course is to study dissipative structure and decay property for hyperbolic systems with dissipation. Consider the following two equations.

$$u_t + Au_x = Bu_{xx} \tag{2}$$

$$u_t + Au_x - Lu = 0 \tag{3}$$

Equation 2 describes the symmetric diffusion process, while symmetric relaxation is modeled by equation 3.

It is always assumed that the matrix A is symmetric. Throughout the course of this lecture three different versions of these two equations are examined in terms of the assumptions that are made on matrices B and L, and what type of additional effects are involved. The simplest case is when one assumes B and L to be symmetric and nonnegative. From this base case we can move in two different directions and arrive at two different versions of these systems. In the first variation, the very same equations are considered with modified conditions, ie. matrices B and L are still assumed to be non-negative, but the symmetry assumption is dropped. In the second variation one considers the original assumptions on symmetry and non-negativity, but the equations are modified with a memory part. More specifically, the equations in the third case are the following.

$$u_t + Au_x = a(t) \star Bu_{xx} \tag{4}$$

$$u_t + Au_x - a(t) \star Lu = 0 \tag{5}$$

During the course the case $a(t) = \mathrm{e}^{-t}$ is considered.

To expound the most important definition, first take the Fourier transforms of equations 2 and 3, and then consider the corresponding eigenvalue problem.

$$(\lambda + I\xi A + \xi^2 B)\phi = 0 \tag{6}$$

$$(\lambda + I\xi A + L)\phi = 0 \tag{7}$$

Equations 2 and 3 are called uniformly dissipative of type (p, q) if

$$\mathrm{Re}(\lambda(I\xi)) \leq \frac{c|\xi|^{2p}}{(1 + |\xi|^2)^q} \tag{8}$$

This concept stands at the center throughout the course and in each of the above mentioned cases we will assert a condition for the system to be uniformly dissipative. In each case this condition will be the Craftmanship Condition, which grasps a very similar meaning in every case, but is naturally defined differently, depending on the concrete equation.

For example, for equation 2 the Craftmanship Condition by definition is satisfied if there exists a skew-symmetric matrix K such that $(KA)^{sy} + B$ is positive definit. Craftmanship Condition in other cases is defined similarly, naturally adapted to the equation.

To continue with our basic case, equation 2, with B being symmetric and non-negative, if Craftmanship Condition is satisfied, then the solution is uniformly dissipative of type $(1, 1)$ and the following bound holds

$$\left|\hat{u}(t,\xi)\right| \leq C\mathrm{e}^{-c\rho(\xi)t}\left|\hat{u}_0(\xi)\right|,$$

where $\rho(\xi) = \frac{\xi^2}{1+\xi^2}$. Moreover, we have the decay estimate

$$\left\|\partial_x^k u(t)\right\|_{L^2} \leq C(1+t)^{-\frac{1}{4}-\frac{k}{2}}\|u_0\|_{L^1} + C\mathrm{e}^{-ct}\left\|\partial_x^k u_0\right\|_{L^2}.$$

It can be shown that for both equations 2 and 3, Craftmanship Condition is sufficient and necessary condition for the system to be uniformly dissipative of $(1,1)$. For the nonsymmetric case of equation 3 the respective Craftmanship Condition is sufficient to be uniformly dissipative.

For the systems with memory-type dissipation, ie. for equations 4 and 5 with a method of introducing a new variable and increasing the dimension of the system we find that memory-type symmetric diffusion is essentially the same as the original relaxation equation with symmetric L matrix, and memory-type symmetric relaxation is essentially the same as the original relaxation equation with non-symmetric L matrix. Craftmanship Conditions can be defined analogously, and the conditions for the respective systems hold and fail equivalently in the diffusion case, for the relaxation system the Craftmanship Condition of the original system is sufficient condition for the Craftmanship Condition of the memory-type system to be satisfied. Analogous results to the above mentioned decay estimate and solution bound can be stated for the non-symmetric and the symmetric memory-type systems as well.

The course explores the relationship between the five different systems defined in the beginning, the Craftmanship Condition, and uniform dissipativity (including decay estimates).

2.3 Advanced Topics on Numerical Methods - Prof. Guglielmi

A-stability, B-stability, Runge-Kutta.

2.4 Introduction to the Incompressible Navier-Stokes equation and the mathematical theory of Turbulence - Prof. Marcati

Euler and Navier-Stokes equations.

3 Seminars

3.1 PDEs Day, 21st March 2017

3.1.1 Viscous vortex rings with axial symmetry - Prof. Gallay

3.1.2 A two phase model with cross and self attractive interactions - Prof. Novaga

3.1.3 Traveling waves for collective movements on a network - Prof. Corli

3.2 School on Uncertainty Quantification for Hyperbolic Equations and Related Topics, 24th-28th April 2017

3.2.1 Uncertainty propagation in transport equations and kinetic polynomials - Prof. Depres

3.2.2 Stochastic asymptotic-preserving methods for multiscale kinetic and quantum transport with uncertainties - Prof. Jin

3.2.3 Kinetic methods for uncertainty propagations in SODEs and SPDEs: from data-driven approximations to functional differential equations - Prof. Venturi

3.2.4 Approximation of stochastic partial differential equations driven by Levy fields - Andreas Barth

3.2.5 Uncertainty qualification in sediment transport shallow flows - M. J. Castro Diaz

3.2.6 Stochastic Galerkin methods for the Boltzmann equation with uncertainty - Jingwei Hu

3.2.7 From uncertain systems of conservation laws to playing with orthonormal polynomial basis - Gael Poette

3.2.8 Structure preserving methods for mean-field equations with random inputs - Mattia Zanella

4 Summer schools

References

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[Evans, 1990] Evans, L. C. (1990). *Weak convergence methods for nonlinear partial differential equations.* Number 74. American Mathematical Soc.

[Majda and Bertozzi, 2002] Majda, A. J. and Bertozzi, A. L. (2002). *Vorticity and incompressible flow*, volume 27. Cambridge University Press.